3

Random vectors in high dimensions

In this chapter we study the distributions of random vectors $X = (X_1, \ldots, X_n) \in \mathbb{R}^n$ where the dimension n is typically very large. Examples of high-dimensional distributions abound in data science. For instance, computational biologists study the expressions of $n \sim 10^4$ genes in the human genome, which can be modeled as a random vector $X = (X_1, \ldots, X_n)$ that encodes the gene expressions of a person randomly drawn from a given population.

Life in high dimensions presents new challenges, which stem from the fact that there is *exponentially more room* in higher dimensions than in lower dimensions. For example, in $\mathbb{R}^n$ the volume of a cube of side 2 is 2^n times larger than the volume of a unit cube, even though the sides of the cubes are just a factor 2 apart (see Figure 3.1). The abundance of room in higher dimensions makes many algorithmic tasks exponentially more difficult, a phenomenon known as the *"curse of dimensionality"*.

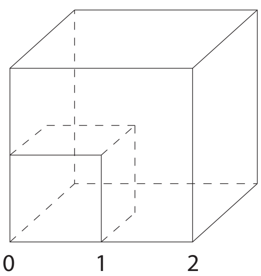

Figure 3.1 The abundance of room in high dimensions: the larger cube has volume exponentially larger than the smaller cube.

Probability in high dimensions offers an array of tools to circumvent these difficulties; some examples will be given in this chapter. We start by examining the Euclidean norm $\|X\|_2$ of a random vector X with independent coordinates, and we show in Section 3.1 that the norm concentrates tightly about its mean. Further basic results and examples of high-dimensional distributions (multivariate normal, spherical, Bernoulli, frames, etc.) are covered in Section 3.2, which also discusses principal component analysis, a powerful data exploratory procedure.

In Section 3.5 we give a probabilistic proof of the classical Grothendieck's inequality, and give an application to semidefinite optimization. We show that one can sometimes relax hard optimization problems to tractable, semidefinite programs, and use Grothendieck's inequality to analyze the quality of such relaxations. In Section 3.6 we give a remarkable example of a semidefinite relaxation of a hard optimization problem – finding the maximum cut of a given graph. We present there the classical Goemans-Williamson randomized approximation algorithm for the maximum cut problem. In Section 3.7 we give an alternative proof of Grothendieck's inequality (and with almost the best known constant) by introducing the kernel trick, a method that has significant applications in machine learning.

3.1 Concentration of the norm

Where in the space $\mathbb{R}^n$ is a random vector $X = (X_1, \ldots, X_n)$ likely to be located? Assume the coordinates X_i are independent random variables with zero means and unit variances. What length do we expect X to have? We have

$$\mathbb{E}\,\|X\|_2^2 = \mathbb{E} \sum_{i=1}^n X_i^2 = \sum_{i=1}^n \mathbb{E}\, X_i^2 = n.$$

So we should expect the length of X to be

$$\|X\|_2 \approx \sqrt{n}.$$

We will see now that X is indeed very close to $\sqrt{n}$ with high probability.

Theorem 3.1.1 (Concentration of the norm). *Let $X = (X_1, \ldots, X_n) \in \mathbb{R}^n$ be a random vector with independent, sub-gaussian coordinates X_i that satisfy $\mathbb{E}\, X_i^2 = 1$. Then*

$$\Big\| \|X\|_2 - \sqrt{n} \Big\|_{\psi_2} \le CK^2,$$

where $K = \max_i \|X_i\|_{\psi_2}$ and C is an absolute constant.[1]

Proof For simplicity, we assume that $K \ge 1$. (Argue that you can make this assumption.) We shall apply Bernstein's deviation inequality for the normalized sum of independent, mean zero random variables

$$\frac{1}{n}\|X\|_2^2 - 1 = \frac{1}{n}\sum_{i=1}^n (X_i^2 - 1).$$

Since the random variable X_i is sub-gaussian, $X_i^2 - 1$ is sub-exponential, and more precisely

$$\begin{aligned} \|X_i^2 - 1\|_{\psi_1} &\le C\|X_i^2\|_{\psi_1} \quad \text{(by centering, see Exercise 2.7.10)} \\ &= C\|X_i\|_{\psi_2}^2 \quad \text{(by Lemma 2.7.6)} \\ &\le CK^2. \end{aligned}$$

[1] From now on, we will always denote various positive absolute constants by C, c, C_1, c_1 without saying this explicitly.

Applying Bernstein's inequality (Corollary 2.8.3), we obtain for any $u \geq 0$ that

$$\mathbb{P}\left\{\left|\frac{1}{n}\|X\|_2^2 - 1\right| \geq u\right\} \leq 2\exp\left(-\frac{cn}{K^4}\min(u^2, u)\right). \tag{3.1}$$

(Here we used that $K^4 \geq K^2$ since we assumed that $K \geq 1$.)

This is a good concentration inequality for $\|X\|_2^2$, from which we are going to deduce a concentration inequality for $\|X\|_2$. To make the link, we can use the following elementary observation that is valid for all numbers $z \geq 0$:

$$|z-1| \geq \delta \quad \text{implies} \quad |z^2 - 1| \geq \max(\delta, \delta^2). \tag{3.2}$$

(Check it!) We obtain for any $\delta \geq 0$ that

$$\begin{aligned}\mathbb{P}\left\{\left|\frac{1}{\sqrt{n}}\|X\|_2 - 1\right| \geq \delta\right\} &\leq \mathbb{P}\left\{\left|\frac{1}{n}\|X\|_2^2 - 1\right| \geq \max(\delta, \delta^2)\right\} \quad \text{(by (3.2))} \\ &\leq 2\exp\left(-\frac{cn}{K^4}\cdot\delta^2\right) \quad \text{(by (3.1) for } u = \max(\delta, \delta^2)).\end{aligned}$$

Changing variables to $t = \delta\sqrt{n}$, we obtain the desired sub-gaussian tail

$$\mathbb{P}\left\{\left|\|X\|_2 - \sqrt{n}\right| \geq t\right\} \leq 2\exp\left(-\frac{ct^2}{K^4}\right) \quad \text{for all } t \geq 0. \tag{3.3}$$

As we know from Section 2.5.2, this is equivalent to the conclusion of the theorem. □

Remark 3.1.2 (Deviation)**.** Theorem 3.1.1 states that with high probability, X takes values very close to the sphere of radius $\sqrt{n}$. In particular, with high probability (say, 0.99), X even stays within *constant distance* from that sphere. Such small, constant deviations could be surprising at the first sight, so let us explain this intuitively. The square of the norm, $S_n := \|X\|_2^2$ has mean n and standard deviation $O(\sqrt{n})$. (Why?) Thus $\|X\|_2 = \sqrt{S_n}$ ought to deviate by $O(1)$ around $\sqrt{n}$. This is because

$$\sqrt{n \pm O(\sqrt{n})} = \sqrt{n} \pm O(1);$$

see Figure 3.2 for an illustration.

Remark 3.1.3 (Anisotropic distributions)**.** After we develop more tools, we will prove a generalization of Theorem 3.1.1 for *anisotropic* random vectors X; see Theorem 6.3.2.

Exercise 3.1.4 (Expectation of the norm)**.** ☕☕☕

(a) Deduce from Theorem 3.1.1 that

$$\sqrt{n} - CK^2 \leq \mathbb{E}\,\|X\|_2 \leq \sqrt{n} + CK^2.$$

(b) Can CK^2 be replaced by $o(1)$, a quantity that vanishes as $n \to \infty$?

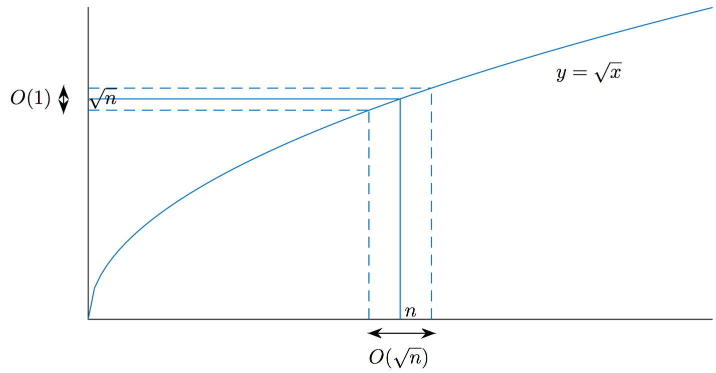

Figure 3.2 Concentration of the norm of a random vector X in $\mathbb{R}^n$. While $\|X\|_2^2$ deviates by $O(\sqrt{n})$ around n, $\|X\|_2$ deviates by $O(1)$ around $\sqrt{n}$.

Exercise 3.1.5 (Variance of the norm)**.** ☕☕☕ Deduce from Theorem 3.1.1 that

$$\mathrm{Var}(\|X\|_2) \le CK^4.$$

Hint: Use Exercise 3.1.4.

The result of the last exercise actually holds not only for sub-gaussian distributions, but for all distributions with bounded fourth moment:

Exercise 3.1.6 (Variance of the norm under finite moment assumptions)**.** ☕☕☕ Let $X = (X_1, \ldots, X_n) \in \mathbb{R}^n$ be a random vector with independent coordinates X_i that satisfy $\mathbb{E}\, X_i^2 = 1$ and $\mathbb{E}\, X_i^4 \le K^4$. Show that

$$\mathrm{Var}(\|X\|_2) \le CK^4.$$

Hint: First check that $\mathbb{E}(\|X\|_2^2 - n)^2 \le K^4 n$ by expansion. This yields in a simple way that $\mathbb{E}(\|X\|_2 - \sqrt{n})^2 \le K^4$. Finally, replace $\sqrt{n}$ by $\mathbb{E}\,\|X\|_2$ arguing like in Exercise 3.1.4.

Exercise 3.1.7 (Small ball probabilities)**.** ☕☕ Let $X = (X_1, \ldots, X_n) \in \mathbb{R}^n$ be a random vector with independent coordinates X_i with continuous distributions. Assume that the densities of X_i are uniformly bounded by 1. Show that, for any $\varepsilon > 0$, we have

$$\mathbb{P}\left\{\|X\|_2 \le \varepsilon\sqrt{n}\right\} \le (C\varepsilon)^n.$$

Hint: While this inequality does not follow from the result of Exercise 2.2.10 (why?), you can prove it by a similar argument.

3.2 Covariance matrices and principal component analysis

In the last section we considered a special class of random variables, those with independent coordinates. Before we study more general situations, let us recall a few basic notions about high-dimensional distributions, which the reader may have already seen in basic courses.

The concept of the *mean* of a random variable generalizes in a straightforward way for a random vectors X taking values in $\mathbb{R}^n$. The notion of variance is replaced in high dimensions by the *covariance matrix* of a random vector $X \in \mathbb{R}^n$, defined as follows:

$$\mathrm{cov}(X) = \mathbb{E}(X-\mu)(X-\mu)^\mathsf{T} = \mathbb{E}\, XX^\mathsf{T} - \mu\mu^\mathsf{T}, \quad \text{where } \mu = \mathbb{E}\, X.$$

Thus $\mathrm{cov}(X)$ is an $n \times n$, symmetric positive semidefinite matrix. The formula for covariance is a direct high-dimensional generalization of the definition of variance for a random variables Z, which is

$$\mathrm{Var}(Z) = \mathbb{E}(Z-\mu)^2 = \mathbb{E}\, Z^2 - \mu^2, \quad \text{where } \mu = \mathbb{E}\, Z.$$

The entries of $\mathrm{cov}(X)$ are the *covariances* of the pairs of coordinates of $X = (X_1, \ldots, X_n)$:

$$\mathrm{cov}(X)_{ij} = \mathbb{E}(X_i - \mathbb{E}\, X_i)(X_j - \mathbb{E}\, X_j).$$

It is sometimes useful to consider the *second moment matrix* of a random vector X, defined as

$$\Sigma = \Sigma(X) = \mathbb{E}\, XX^\mathsf{T}.$$

The second moment matrix is a higher dimensional generalization of the second moment $\mathbb{E}\, Z^2$ of a random variable Z. By translation (replacing X with $X - \mu$), we can assume in many problems that X has zero mean, and thus covariance and second moment matrices are equal:

$$\mathrm{cov}(X) = \Sigma(X).$$

This observation allows us to mostly focus on the second moment matrix $\Sigma = \Sigma(X)$ rather than on the covariance $\mathrm{cov}(X)$ in the future.

Like the covariance matrix, the second moment matrix Σ is also an $n \times n$, symmetric and positive semidefinite matrix. The spectral theorem for such matrices says that all eigenvalues s_i of Σ are real and non-negative. Moreover, Σ can be expressed via spectral decomposition as

$$\Sigma = \sum_{i=1}^{n} s_i u_i u_i^\mathsf{T},$$

where $u_i \in \mathbb{R}^n$ are the eigenvectors of Σ. We usually arrange the terms in this sum so that the eigenvalues s_i are decreasing.

3.2.1 Principal component analysis

The spectral decomposition of Σ is of utmost importance in applications where the distribution of a random vector X in $\mathbb{R}^n$ represents data, for example the genetic data we mentioned on p. 41. The eigenvector u_1 corresponding to the largest eigenvalue s_1 defines the first *principal direction*. This is the direction in which the distribution is most extended, and it explains most of the variability in the data. The next eigenvector u_2 (corresponding to the next largest eigenvalue s_2) defines the next principal direction; it best explains the remaining variations in the data, and so on. This is illustrated in the Figure 3.3.

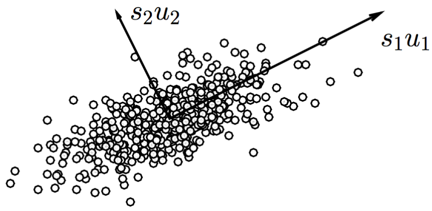

Figure 3.3 Illustration of the PCA. 200 sample points are shown from a distribution in $\mathbb{R}^2$. The covariance matrix Σ has eigenvalues s_i and eigenvectors u_i.

It often happens with real data that only a few eigenvalues s_i are large and can be considered as informative; the remaining eigenvalues are small and considered as noise. In such situations, a few principal directions can explain most variability in the data. Even though the data is presented in a high-dimensional space $\mathbb{R}^n$, such data is essentially *low dimensional*. It clusters near the low-dimensional subspace E spanned by the first few principal components.

The most basic data analysis algorithm, called the *principal component analysis* (PCA), computes the first few principal components and then projects the data in $\mathbb{R}^n$ onto the subspace E spanned by them. This considerably reduces the dimension of the data and simplifies the data analysis. For example, if E is two- or three-dimensional, PCA allows one to visualize the data.

3.2.2 Isotropy

We might remember from a basic probability course how it is often convenient to assume that random variables have zero means and unit variances. This is also true in higher dimensions, where the notion of isotropy generalizes the assumption of unit variance.

Definition 3.2.1 (Isotropic random vectors)**.** A random vector X in $\mathbb{R}^n$ is called *isotropic* if

$$\Sigma(X) = \mathbb{E}\, X X^{\mathsf{T}} = I_n$$

where I_n denotes the identity matrix in $\mathbb{R}^n$.

Recall that any random variable X with positive variance can be reduced by translation and dilation to the *standard score* – a random variable Z with zero mean and unit variance, namely

$$Z = \frac{X - \mu}{\sqrt{\mathrm{Var}(X)}}.$$

The following exercise gives a high-dimensional version of standard score.

Exercise 3.2.2 (Reduction to isotropy). ☕

(a) Let Z be a mean zero, isotropic random vector in $\mathbb{R}^n$. Let $\mu \in \mathbb{R}^n$ be a fixed vector and Σ be a fixed $n \times n$ symmetric positive semidefinite matrix. Check that the random vector

$$X := \mu + \Sigma^{1/2} Z$$

has mean μ and covariance matrix $\mathrm{cov}(X) = \Sigma$.

(b) Let X be a random vector with mean μ and invertible covariance matrix $\Sigma = \mathrm{cov}(X)$. Check that the random vector

$$Z := \Sigma^{-1/2}(X - \mu)$$

is an isotropic, mean zero random vector.

This observation will allow us in many future results about random vectors to assume without loss of generality that they have zero means and are isotropic.

3.2.3 Properties of isotropic distributions

Lemma 3.2.3 (Characterization of isotropy). *A random vector X in $\mathbb{R}^n$ is isotropic if and only if*

$$\mathbb{E}\,\langle X, x\rangle^2 = \|x\|_2^2 \quad \textit{for all } x \in \mathbb{R}^n.$$

Proof Recall that two symmetric $n \times n$ matrices A and B are equal if and only if $x^\mathsf{T} A x = x^\mathsf{T} B x$ for all $x \in \mathbb{R}^n$. (Check this!) Thus X is isotropic if and only if

$$x^\mathsf{T}\left(\mathbb{E}\, X X^\mathsf{T}\right) x = x^\mathsf{T} I_n x \quad \text{for all } x \in \mathbb{R}^n.$$

The left side of this identity equals $\mathbb{E}\,\langle X, x\rangle^2$ and the right side is $\|x\|_2^2$. This completes the proof. □

If x is a unit vector in Lemma 3.2.3, we can view $\langle X, x\rangle$ as a one-dimensional marginal of the distribution of X, obtained by projecting X onto the direction of x. Then a mean-zero random vector X is isotropic if and only if *all one-dimensional marginals of X have unit variance.* Informally, this means that an isotropic distribution is extended evenly in all directions.

Lemma 3.2.4. *Let X be an isotropic random vector in $\mathbb{R}^n$. Then*

$$\mathbb{E}\,\|X\|_2^2 = n.$$

Moreover, if X and Y are two independent isotropic random vectors in $\mathbb{R}^n$, then

$$\mathbb{E}\,\langle X, Y\rangle^2 = n.$$

Proof To prove the first part, we have

$$\begin{aligned}\mathbb{E}\,\|X\|_2^2 &= \mathbb{E}\,X^\mathsf{T}X = \mathbb{E}\,\mathrm{tr}(X^\mathsf{T}X) \quad \text{(viewing } X^\mathsf{T}X \text{ as a } 1\times 1 \text{ matrix)}\\ &= \mathbb{E}\,\mathrm{tr}(XX^\mathsf{T}) \quad \text{(by the cyclic property of trace)}\\ &= \mathrm{tr}(\mathbb{E}\,XX^\mathsf{T}) \quad \text{(by linearity)}\\ &= \mathrm{tr}(I_n) \quad \text{(by isotropy)}\\ &= n.\end{aligned}$$

To prove the second part, we use a conditioning argument. Fix a realization of Y and take the conditional expectation (with respect to X) which we denote $\mathbb{E}_X$. The law of total expectation says that

$$\mathbb{E}\,\langle X,Y\rangle^2 = \mathbb{E}_Y\,\mathbb{E}_X\left[\langle X,Y\rangle^2 \mid Y\right],$$

where by $\mathbb{E}_Y$ we of course denote the expectation with respect to Y. To compute the inner expectation, we apply Lemma 3.2.3 with $x = Y$ and conclude that the inner expectation equals $\|Y\|_2^2$. Thus

$$\begin{aligned}\mathbb{E}\,\langle X,Y\rangle^2 &= \mathbb{E}_Y\,\|Y\|_2^2\\ &= n \quad \text{(by the first part of lemma).}\end{aligned}$$

The proof is complete. □

Remark 3.2.5 (Almost orthogonality of independent vectors)**.** Let us normalize the random vectors X and Y in Lemma 3.2.4, setting

$$\overline{X} := \frac{X}{\|X\|_2} \quad \text{and} \quad \overline{Y} := \frac{Y}{\|Y\|_2}.$$

Lemma 3.2.4 is basically telling us that[2] $\|X\|_2 \asymp \sqrt{n}$, $\|Y\|_2 \asymp \sqrt{n}$ and $\langle X,Y\rangle \asymp \sqrt{n}$ with high probability, which implies that

$$\left|\left\langle \overline{X},\overline{Y}\right\rangle\right| \asymp \frac{1}{\sqrt{n}}$$

Thus, in high-dimensional spaces independent and isotropic random vectors tend to be *almost orthogonal*, see Figure 3.4.

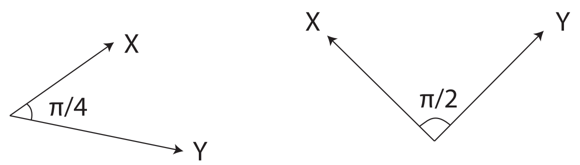

Figure 3.4 Independent isotropic random vectors tend to be almost orthogonal in high dimensions but not in low dimensions. On the plane, the average angle is $\pi/4$, while in high dimensions it is close to $\pi/2$.

[2] This argument is not entirely rigorous, since Lemma 3.2.4 is about expectation and not high probability. To make it more rigorous, one can use Theorem 3.1.1 about concentration of the norm.

This may sound surprising since this is not the case in low dimensions. For example, the angle between two random independent and uniformly distributed directions on the plane has mean $\pi/4$. (Check!) But in higher dimensions, there is much more room as we mentioned in the beginning of this chapter. This is an intuitive reason why random directions in high-dimensional spaces tend to be very far from each other, i.e. almost orthogonal.

Exercise 3.2.6 (Distance between independent isotropic vectors). Let X and Y be independent, mean zero, isotropic random vectors in $\mathbb{R}^n$. Check that

$$\mathbb{E}\,\|X - Y\|_2^2 = 2n.$$

3.3 Examples of high-dimensional distributions

In this section we give several basic examples of isotropic high-dimensional distributions.

3.3.1 Spherical and Bernoulli distributions

The coordinates of an isotropic random vector are always uncorrelated (why?), but they are not necessarily independent. An example of this situation is the *spherical distribution*, where a random vector X is uniformly distributed[3] on the Euclidean sphere in $\mathbb{R}^n$ with center at the origin and radius $\sqrt{n}$:

$$X \sim \mathrm{Unif}\left(\sqrt{n}\, S^{n-1}\right).$$

Exercise 3.3.1. Show that the spherically distributed random vector X is isotropic. Argue that the coordinates of X are not independent.

A good example of a discrete isotropic distribution in $\mathbb{R}^n$ is the *symmetric Bernoulli* distribution. We say that a random vector $X = (X_1, \ldots, X_n)$ is symmetric Bernoulli if the coordinates X_i are independent, symmetric Bernoulli random variables. Equivalently, we may say that X is uniformly distributed on the unit discrete cube in $\mathbb{R}^n$:

$$X \sim \mathrm{Unif}\left(\{-1, 1\}^n\right).$$

The symmetric Bernoulli distribution is isotropic. (Check!)

More generally, we may consider any random vector $X = (X_1, \ldots, X_n)$ whose coordinates X_i are independent random variables with zero mean and unit variance. Then X is an isotropic vector in $\mathbb{R}^n$. (Why?)

[3] More rigorously, we say that X is uniformly distributed on $\sqrt{n}\, S^{n-1}$ if, for every (Borel) subset $E \subset S^{n-1}$, the probability $\mathbb{P}\left\{X \in E\right\}$ equals the ratio of the $(n-1)$-dimensional areas of E and S^{n-1}.

3.3.2 Multivariate normal

One of the most important high-dimensional distributions is Gaussian, or multivariate normal. From a basic probability course we know that a random vector $g = (g_1, \ldots, g_n)$ has the *standard normal distribution* in $\mathbb{R}^n$, denoted

$$g \sim N(0, I_n),$$

if the coordinates g_i are independent standard normal random variables $N(0,1)$. The density of Z is then the product of the n standard normal densities (1.6), which is

$$f(x) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-x_i^2/2} = \frac{1}{(2\pi)^{n/2}} e^{-\|x\|_2^2/2}, \quad x \in \mathbb{R}^n. \tag{3.4}$$

The standard normal distribution is isotropic. (Why?)

Note that the standard normal density (3.4) is *rotation invariant*, since $f(x)$ depends only on the length but not the direction of x. We can equivalently express this observation as follows:

Proposition 3.3.2 (Rotation invariance). *Consider a random vector $g \sim N(0, I_n)$ and a fixed orthogonal matrix U. Then*

$$Ug \sim N(0, I_n).$$

Exercise 3.3.3 (Rotation invariance). Deduce the following properties from the rotation invariance of the normal distribution.

(a) Consider a random vector $g \sim N(0, I_n)$ and a fixed vector $u \in \mathbb{R}^n$. Then

$$\langle g, u \rangle \sim N(0, \|u\|_2^2).$$

(b) Consider independent random variables $X_i \sim N(0, \sigma_i^2)$. Then

$$\sum_{i=1}^n X_i \sim N(0, \sigma^2) \quad \text{where} \quad \sigma^2 = \sum_{i=1}^n \sigma_i^2.$$

(c) Let G be an $m \times n$ Gaussian random matrix, i.e. the entries of G are independent $N(0,1)$ random variables. Let $u \in \mathbb{R}^n$ be a fixed unit vector. Then

$$Gu \sim N(0, I_m).$$

Let us also recall the notion of the *general* normal distribution $N(\mu, \Sigma)$. Consider a vector $\mu \in \mathbb{R}^n$ and an invertible $n \times n$ positive semidefinite matrix Σ. According to Exercise 3.2.2, the random vector $X := \mu + \Sigma^{1/2} Z$ has mean μ and covariance matrix $\Sigma(X) = \Sigma$. Such X is said to have a general normal distribution in $\mathbb{R}^n$, denoted

$$X \sim N(\mu, \Sigma).$$

Summarizing, we have $X \sim N(\mu, \Sigma)$ if and only if

$$Z := \Sigma^{-1/2}(X - \mu) \sim N(0, I_n).$$

The density of $X \sim N(\mu, \Sigma)$ can be computed by the change of variables formula, and it equals

$$f_X(x) = \frac{1}{(2\pi)^{n/2}\det(\Sigma)^{1/2}}\, e^{-(x-\mu)^\mathsf{T}\Sigma^{-1}(x-\mu)/2}, \quad x \in \mathbb{R}^n. \tag{3.5}$$

Figure 3.5 shows examples of two densities of multivariate normal distributions.

An important observation is that the coordinates of a random vector $X \sim N(\mu, \Sigma)$ are independent if and only if they are uncorrelated. (In this case $\Sigma = I_n$.)

Exercise 3.3.4 (Characterization of normal distribution)**.** ☕☕☕ Let X be a random vector in $\mathbb{R}^n$. Show that X has a multivariate normal distribution if and only if every one-dimensional marginal $\langle X, \theta\rangle$, $\theta \in \mathbb{R}^n$, has a (univariate) normal distribution.

Hint: Utilize a version of Cramér-Wold's theorem, which states that the totality of the distributions of one-dimensional marginals determine the distribution in $\mathbb{R}^n$ uniquely. More precisely, if X and Y are random vectors in $\mathbb{R}^n$ such that $\langle X, \theta\rangle$ and $\langle Y, \theta\rangle$ have the same distribution for each $\theta \in \mathbb{R}^n$, then X and Y have the same distribution.

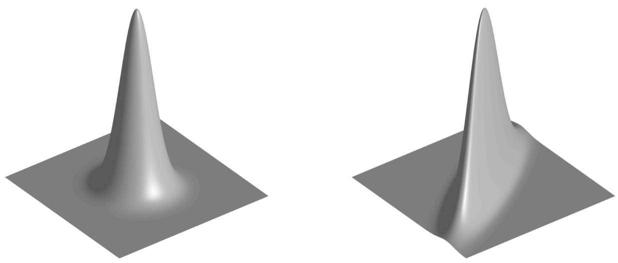

Figure 3.5 The densities of the isotropic distribution $N(0, I_2)$ and a non-isotropic distribution $N(0, \Sigma)$.

Exercise 3.3.5. ☕ Let $X \sim N(0, I_n)$.

(a) Show that, for any fixed vectors $u, v \in \mathbb{R}^n$, we have

$$\mathbb{E}\,\langle X, u\rangle\,\langle X, v\rangle = \langle u, v\rangle\,. \tag{3.6}$$

(b) Given a vector $u \in \mathbb{R}^n$, consider the random variable $X_u := \langle X, u\rangle$. From Exercise 3.3.3 we know that $X_u \sim N(0, \|u\|_2^2)$. Check that

$$\|X_u - X_v\|_{L^2} = \|u - v\|_2$$

for any fixed vectors $u, v \in \mathbb{R}^n$. (Here $\|\cdot\|_{L^2}$ denotes the norm in the Hilbert space L^2 of random variables, which we introduced in (1.1).)

Exercise 3.3.6. ☕ Let G be an $m \times n$ Gaussian random matrix, i.e. the entries of G are independent $N(0,1)$ random variables. Let $u, v \in \mathbb{R}^n$ be unit orthogonal vectors. Prove that Gu and Gv are independent $N(0, I_m)$ random vectors.

Hint: Reduce the problem to the case where u and v are collinear with canonical basis vectors of $\mathbb{R}^n$.

3.3.3 Similarity of normal and spherical distributions

Contradicting our low dimensional intuition, the standard normal distribution $N(0, I_n)$ in high dimensions is *not* concentrated close to the origin, where the density is maximal. Instead, it is concentrated *in a thin spherical shell* around the sphere of radius $\sqrt{n}$, a shell of width $O(1)$. Indeed, the concentration inequality (3.3) for the norm of $g \sim N(0, I_n)$ states that

$$\mathbb{P}\left\{\left|\|g\|_2 - \sqrt{n}\right| \geq t\right\} \leq 2\exp(-ct^2) \quad \text{for all } t \geq 0. \tag{3.7}$$

This observation suggests that the normal distribution should be quite similar to the uniform distribution on the sphere. Let us clarify the relation.

Exercise 3.3.7 (Normal and spherical distributions)**.** ☕ Let us represent $g \sim N(0, I_n)$ in polar form as

$$g = r\theta$$

where $r = \|g\|_2$ is the length and $\theta = g/\|g\|_2$ is the direction of g. Prove the following:

(a) The length r and direction θ are independent random variables.
(b) The direction θ is uniformly distributed on the unit sphere S^{n-1}.

Concentration inequality (3.7) says that $r = \|g\|_2 \approx \sqrt{n}$ with high probability, so

$$g \approx \sqrt{n}\,\theta \sim \mathrm{Unif}\left(\sqrt{n}S^{n-1}\right).$$

In other words, the standard normal distribution in high dimensions is close to the uniform distribution on the sphere of radius $\sqrt{n}$, i.e.

$$N(0, I_n) \approx \mathrm{Unif}\left(\sqrt{n}S^{n-1}\right). \tag{3.8}$$

Figure 3.6 illustrates this fact that goes against our intuition that has been trained in low dimensions.

3.3.4 Frames

For an example of an extremely discrete distribution, consider a *coordinate random vector* X uniformly distributed in the set $\{\sqrt{n}\,e_i\}_{i=1}^n$ where $\{e_i\}_{i=1}^n$ is the canonical basis of $\mathbb{R}^n$:

$$X \sim \mathrm{Unif}\left\{\sqrt{n}\,e_i : \ i = 1, \ldots, n\right\}.$$

Then X is an isotropic random vector in $\mathbb{R}^n$. (Check!)

Of all high-dimensional distributions, Gaussian is often the most convenient to prove results for, so we may think of it as "the best" distribution. The coordinate distribution, the most discrete of all distributions, is "the worst".

A general class of discrete, isotropic distributions arises in the area of signal processing under the name of *frames*.

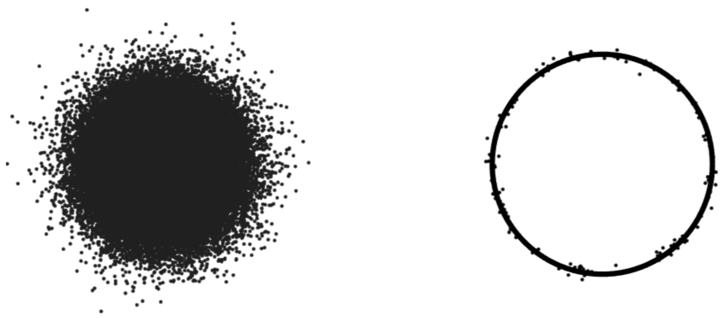

Figure 3.6 A Gaussian point cloud in two dimensions (left) and its intuitive visualization in high dimensions (right). In high dimensions, the standard normal distribution is very close to the uniform distribution on the sphere of radius $\sqrt{n}$.

Definition 3.3.8. A *frame* is a set of vectors $\{u_i\}_{i=1}^N$ in $\mathbb{R}^n$ which obeys an approximate Parseval's identity, i.e. there exist numbers $A, B > 0$ called *frame bounds* such that

$$A\|x\|_2^2 \leq \sum_{i=1}^N \langle u_i, x\rangle^2 \leq B\|x\|_2^2 \quad \text{for all } x \in \mathbb{R}^n.$$

If $A = B$ the set $\{u_i\}_{i=1}^N$ is called a *tight frame*.

Exercise 3.3.9. Show that $\{u_i\}_{i=1}^N$ is a tight frame in $\mathbb{R}^n$ with bound A if and only if

$$\sum_{i=1}^N u_i u_i^\mathsf{T} = A I_n. \tag{3.9}$$

Hint: Proceed similarly to the proof of Lemma 3.2.3.

Multiplying both sides of (3.9) by a vector x, we see that

$$\sum_{i=1}^N \langle u_i, x\rangle\, u_i = Ax \quad \text{for any } x \in \mathbb{R}^n. \tag{3.10}$$

This is a *frame expansion* of a vector x, and it should look familiar. Indeed, if $\{u_i\}$ is an orthonormal basis, then (3.10) is just a classical basis expansion of x, and it holds with $A = 1$.

We can think of tight frames as generalizations of orthogonal bases *without the linear independence* requirement. Any orthonormal basis in $\mathbb{R}^n$ is clearly a tight frame. But so is the "Mercedez-Benz frame", a set of three equidistant points on a circle in $\mathbb{R}^2$ shown on Figure 3.7.

Now we are ready to connect the concept of frames to probability. We show that tight frames correspond to isotropic distributions, and vice versa.

Lemma 3.3.10 (Tight frames and isotropic distributions). *(a) Consider a tight*

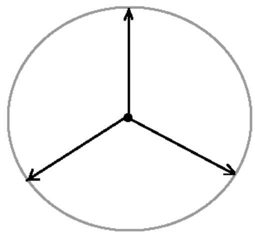

Figure 3.7 the Mercedes-Benz frame. A set of equidistant points on the circle form a tight frame in $\mathbb{R}^2$.

frame $\{u_i\}_{i=1}^N$ in $\mathbb{R}^n$ with frame bounds $A = B$. Let X be a random vector that is uniformly distributed in the set of frame elements, i.e.

$$X \sim \mathrm{Unif}\{u_i : \ i = 1, \ldots, N\}.$$

Then $(N/A)^{1/2}X$ is an isotropic random vector in $\mathbb{R}^n$.

(b) Consider an isotropic random vector X in $\mathbb{R}^n$ that takes a finite set of values x_i with probabilities p_i each, $i = 1, \ldots, N$. Then the vectors

$$u_i := \sqrt{p_i}\, x_i, \quad i = 1, \ldots, N,$$

form a tight frame in $\mathbb{R}^n$ with bounds $A = B = 1$.

Proof 1. Without loss of generality, we can assume that $A = N$. (Why?) The assumptions and (3.9) imply that

$$\sum_{i=1}^N u_i u_i^\mathsf{T} = N I_n.$$

Dividing both sides by N and interpreting $\frac{1}{N}\sum_{i=1}^N$ as an expectation, we conclude that X is isotropic.

2. Isotropy of X means that

$$\mathbb{E}\, X X^\mathsf{T} = \sum_{i=1}^N p_i x_i x_i^\mathsf{T} = I_n.$$

Denoting $u_i := \sqrt{p_i}\, x_i$, we obtain (3.9) with $A = 1$. $\square$

3.3.5 Isotropic convex sets

Our last example of a high-dimensional distribution comes from convex geometry. Consider a bounded convex set K in $\mathbb{R}^n$ with non-empty interior; such sets are called *convex bodies*. Let X be a random vector uniformly distributed in K, according to the probability measure given by normalized volume in K:

$$X \sim \mathrm{Unif}(K).$$

Assume that $\mathbb{E} X = 0$ (translate K appropriately to achieve this) and denote the covariance matrix of X by Σ. Then by Exercise 3.2.2, the random vector $Z := \Sigma^{-1/2} X$ is isotropic. Note that Z is uniformly distributed in the linearly transformed copy of K:

$$Z \sim \mathrm{Unif}(\Sigma^{-1/2} K).$$

(Why?) Summarizing, we found a linear transformation $T := \Sigma^{-1/2}$ which makes the uniform distribution on TK isotropic. The body TK is sometimes called isotropic itself.

In algorithmic convex geometry, one can think of the isotropic convex body TK as a *well conditioned* version of K, with T playing the role of a pre-conditioner, see Figure 3.8. Algorithms related to convex bodies K (such as computing the volume of K) tend to work better for well-conditioned bodies K.

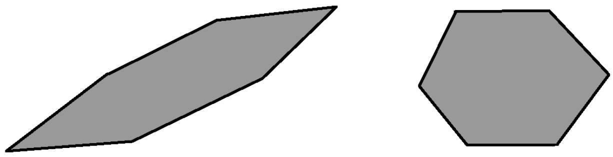

Figure 3.8 A convex body K on the left is transformed into an isotropic convex body TK on the right. The pre-conditioner T is computed from the covariance matrix Σ of K as $T = \Sigma^{-1/2}$.

3.4 Sub-gaussian distributions in higher dimensions

The concept of sub-gaussian distributions, which we introduced in Section 2.5, can be extended to higher dimensions. To see how, recall from Exercise 3.3.4 that the multivariate normal distribution can be characterized through its *one-dimensional marginals*, or projections onto lines: a random vector X has a normal distribution in $\mathbb{R}^n$ if and only if the one-dimensional marginals $\langle X, x \rangle$ are normal for all $x \in \mathbb{R}^n$. Guided by this characterization, it is natural to define multivariate sub-gaussian distributions as follows.

Definition 3.4.1 (Sub-gaussian random vectors)**.** A random vector X in $\mathbb{R}^n$ is called *sub-gaussian* if the one-dimensional marginals $\langle X, x \rangle$ are sub-gaussian random variables for all $x \in \mathbb{R}^n$. The *sub-gaussian norm* of X is defined as

$$\|X\|_{\psi_2} = \sup_{x \in S^{n-1}} \| \langle X, x \rangle \|_{\psi_2}.$$

A good example of a sub-gaussian random vector is a random vector with independent, sub-gaussian coordinates:

Lemma 3.4.2 (Sub-gaussian distributions with independent coordinates)**.** *Let*

$X = (X_1, \ldots, X_n) \in \mathbb{R}^n$ be a random vector with independent, mean zero, sub-gaussian coordinates X_i. Then X is a sub-gaussian random vector, and

$$\|X\|_{\psi_2} \le C \max_{i \le n} \|X_i\|_{\psi_2}.$$

Proof This is an easy consequence of the fact that a sum of independent sub-gaussian random variables is sub-gaussian, which we proved in Proposition 2.6.1. Indeed, for a fixed unit vector $x = (x_1, \ldots, x_n) \in S^{n-1}$ we have

$$\begin{aligned} \| \langle X, x \rangle \|_{\psi_2}^2 &= \Big\| \sum_{i=1}^n x_i X_i \Big\|_{\psi_2}^2 \le C \sum_{i=1}^n x_i^2 \|X_i\|_{\psi_2}^2 \quad \text{(by Proposition 2.6.1)} \\ &\le C \max_{i \le n} \|X_i\|_{\psi_2}^2 \quad \Big(\text{using that } \sum_{i=1}^n x_i^2 = 1\Big). \end{aligned}$$

This completes the proof. □

Exercise 3.4.3. ☕☕ This exercise clarifies the role of independence of coordinates in Lemma 3.4.2.

1. Let $X = (X_1, \ldots, X_n) \in \mathbb{R}^n$ be a random vector with sub-gaussian coordinates X_i. Show that X is a sub-gaussian random vector.
2. Nevertheless, find an example of a random vector X with

$$\|X\|_{\psi_2} \gg \max_{i \le n} \|X_i\|_{\psi_2}.$$

Many important high-dimensional distributions are sub-gaussian, but some are not. We now explore some basic distributions.

3.4.1 Gaussian and Bernoulli distributions

As we already noted, *multivariate normal distribution* $N(\mu, \Sigma)$ is sub-gaussian. Moreover, the standard normal random vector $X \sim N(0, I_n)$ has sub-gaussian norm of order $O(1)$:

$$\|X\|_{\psi_2} \le C.$$

(Indeed, all one-dimensional marginals of X are $N(0,1)$.)

Next, consider the multivariate *symmetric Bernoulli* distribution that we introduced in Section 3.3.1. A random vector X with this distribution has independent, symmetric Bernoulli coordinates, so Lemma 3.4.2 yields that

$$\|X\|_{\psi_2} \le C.$$

3.4.2 Discrete distributions

Let us now pass to discrete distributions. The extreme example we considered in Section 3.3.4 is the *coordinate distribution*. Recall that random vector X with coordinate distribution is uniformly distributed in the set $\{\sqrt{n} e_i : \; i = 1, \ldots, n\}$, where e_i denotes the the n-element set of the canonical basis vectors in $\mathbb{R}^n$.

Is X sub-gaussian? Formally, yes. In fact, every distribution supported in a finite set is sub-gaussian. (Why?) But, unlike Gaussian and Bernoulli distributions, the coordinate distribution has a very large sub-gaussian norm.

Exercise 3.4.4. ☕ Show that

$$\|X\|_{\psi_2} \asymp \sqrt{\frac{n}{\log n}}.$$

Such large norm makes it useless to think of X as a sub-gaussian random vector.

More generally, discrete distributions do not make nice sub-gaussian distributions, unless they are supported on exponentially large sets:

Exercise 3.4.5. ☕☕☕☕ Let X be an isotropic random vector supported in a finite set $T \subset \mathbb{R}^n$. Show that in order for X to be sub-gaussian with $\|X\|_{\psi_2} = O(1)$, the cardinality of the set must be exponentially large in n:

$$|T| \geq e^{cn}.$$

In particular, this observation rules out *frames* (see Section 3.3.4) as good sub-gaussian distributions unless they have exponentially many terms (in which case they are mostly useless in practice).

3.4.3 Uniform distribution on the sphere

In all previous examples, good sub-gaussian random vectors had independent coordinates. This is not necessary. A good example is the uniform distribution on the sphere of radius $\sqrt{n}$, which we discussed in Section 3.3.1. We will show that it is sub-gaussian by reducing it to the Gaussian distribution $N(0, I_n)$.

Theorem 3.4.6 (Uniform distribution on the sphere is sub-gaussian)**.** *Let X be a random vector uniformly distributed on the Euclidean sphere in $\mathbb{R}^n$ with center at the origin and radius $\sqrt{n}$:*

$$X \sim \mathrm{Unif}\left(\sqrt{n}\, S^{n-1}\right).$$

Then X is sub-gaussian, and

$$\|X\|_{\psi_2} \leq C.$$

Proof Consider a standard normal random vector $g \sim N(0, I_n)$. As we noted in Exercise 3.3.7, the direction $g/\|g\|_2$ is uniformly distributed on the unit sphere S^{n-1}. Thus, by rescaling we can represent a random vector $X \sim \mathrm{Unif}\left(\sqrt{n}\, S^{n-1}\right)$ as

$$X = \sqrt{n}\, \frac{g}{\|g\|_2}.$$

We need to show that all one-dimensional marginals $\langle X, x\rangle$ are sub-gaussian.

By rotation invariance, we may assume that $x = (1, 0, \dots, 0)$, in which case $\langle X, x \rangle = X_1$, the first coordinate of X. We want to bound the tail probability

$$p(t) := \mathbb{P}\left\{|X_1| \geq t\right\} = \mathbb{P}\left\{\frac{|g_1|}{\|g\|_2} \geq \frac{t}{\sqrt{n}}\right\}.$$

The concentration of norm (Theorem 3.1.1) implies that

$$\|g\|_2 \approx \sqrt{n} \quad \text{with high probability.}$$

This reduces the problem to bounding $\mathbb{P}\left\{|g_1| \geq t\right\}$, but as we know from (2.3), this tail is sub-gaussian.

Let us do this argument more carefully. Theorem 3.1.1 implies that

$$\Big\| \|g\|_2 - \sqrt{n} \Big\|_{\psi_2} \leq C.$$

Thus the event

$$\mathcal{E} := \left\{\|g\|_2 \geq \frac{\sqrt{n}}{2}\right\}$$

is likely: by (2.14) its complement $\mathcal{E}^c$ has probability

$$\mathbb{P}(\mathcal{E}^c) \leq 2\exp(-cn). \tag{3.11}$$

Then the tail probability can be bounded as follows:

$$\begin{aligned} p(t) &\leq \mathbb{P}\left\{\frac{|g_1|}{\|g\|_2} \geq \frac{t}{\sqrt{n}} \text{ and } \mathcal{E}\right\} + \mathbb{P}(\mathcal{E}^c) \\ &\leq \mathbb{P}\left\{|g_1| \geq \frac{t}{2} \text{ and } \mathcal{E}\right\} + 2\exp(-cn) \quad \text{(by definition of } \mathcal{E} \text{ and (3.11))} \\ &\leq 2\exp(-t^2/8) + 2\exp(-cn) \quad \text{(drop } \mathcal{E} \text{ and use (2.3)).} \end{aligned}$$

Consider two cases. If $t \leq \sqrt{n}$ then $2\exp(-cn) \leq 2\exp(-ct^2/8)$, and we conclude that

$$p(t) \leq 4\exp(-c't^2)$$

as desired. In the opposite case where $t > \sqrt{n}$, the tail probability $p(t) = \mathbb{P}\left\{|X_1| \geq t\right\}$ trivially equals zero, since we always have $|X_1| \leq \|X\|_2 = \sqrt{n}$. This completes the proof by the characterization of sub-gaussian distributions (recall Proposition 2.5.2 and Remark 2.5.3). □

Exercise 3.4.7 (Uniform distribution on the Euclidean ball)**.** ☕☕ Extend Theorem 3.4.6 for the uniform distribution on the Euclidean ball $B(0, \sqrt{n})$ in $\mathbb{R}^n$ centered at the origin and with radius $\sqrt{n}$. Namely, show that a random vector

$$X \sim \text{Unif}\left(B(0, \sqrt{n})\right)$$

is sub-gaussian, and

$$\|X\|_{\psi_2} \leq C.$$

Remark 3.4.8 (Projective limit theorem)**.** Theorem 3.4.6 should be compared to the so-called projective central limit theorem. It states that the marginals of the uniform distribution on the sphere become asymptotically normal as n increases, see Figure 3.9. Precisely, if $X \sim \text{Unif}\left(\sqrt{n}\, S^{n-1}\right)$ then for any fixed unit vector x we have

$$\langle X, x\rangle \to N(0,1) \quad \text{in distribution as } n \to \infty.$$

Thus we can view Theorem 3.4.6 as a concentration version of the Projective Limit Theorem, in the same sense as Hoeffding's inequality in Section 2.2 is a concentration version of the classical central limit theorem.

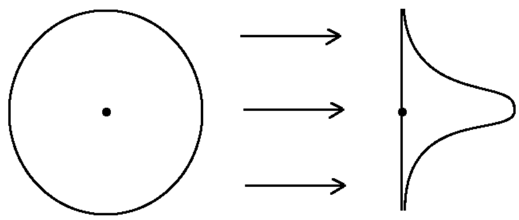

Figure 3.9 The projective central limit theorem: the projection of the uniform distribution on the sphere of radius $\sqrt{n}$ onto a line converges to the normal distribution $N(0,1)$ as $n \to \infty$.

3.4.4 Uniform distribution on convex sets

To conclude this section, let us return to the class of uniform distributions on *convex sets* which we discussed in Section 3.3.5. Let K be a convex body and

$$X \sim \text{Unif}(K)$$

be an isotropic random vector. Is X always sub-gaussian?

For some bodies K, this is the case. Examples include the Euclidean ball of radius $\sqrt{n}$ (by Exercise 3.4.7) and the unit cube $[-1,1]^n$ (according to Lemma 3.4.2). For some other bodies, this is not true:

Exercise 3.4.9. ☕☕☕ Consider a ball of the ℓ_1 norm in $\mathbb{R}^n$:

$$K := \left\{x \in \mathbb{R}^n : \ \|x\|_1 \le r\right\}.$$

(a) Show that the uniform distribution on K is isotropic for some $r \asymp n$.
(b) Show that the subgaussian norm of this distribution is *not* bounded by an absolute constant as the dimension n grows.

Nevertheless, it is possible to prove a weaker result for a general isotropic convex body K. The random vector $X \sim \text{Unif}(K)$ has all *sub-exponential* marginals, and

$$\|\langle X, x\rangle\|_{\psi_1} \le C$$

for all unit vectors x. This result follows from C. Borell's lemma, which itself is a consequence of Brunn-Minkowski inequality; see [81, Section 2.2.b$_3$].

Exercise 3.4.10. ☕☕ Show that the concentration inequality in Theorem 3.1.1 may not hold for a general isotropic sub-gaussian random vector X. Thus, independence of the coordinates of X is an essential requirement in that result.

3.5 Application: Grothendieck's inequality and semidefinite programming

In this and the next section, we use high-dimensional Gaussian distributions to pursue some problems that have seemingly nothing to do with probability. Here we give a probabilistic proof of Grothendieck's inequality, a remarkable result which we will use later in the analysis of some computationally hard problems.

Theorem 3.5.1 (Grothendieck's inequality)**.** *Consider an $m \times n$ matrix (a_{ij}) of real numbers. Assume that, for any numbers $x_i, y_j \in \{-1, 1\}$, we have*

$$\Big| \sum_{i,j} a_{ij} x_i y_j \Big| \le 1.$$

Then, for any Hilbert space H and any vectors $u_i, v_j \in H$ satisfying $\|u_i\| = \|v_j\| = 1$, we have

$$\Big| \sum_{i,j} a_{ij} \langle u_i, v_j \rangle \Big| \le K,$$

where $K \le 1.783$ is an absolute constant.

There is apparently nothing random in the statement of this theorem, but our proof of this result will be probabilistic. We will actually give two proofs of Grothendieck's inequality. The one given in this section will yield a much worse bound on the constant K, namely $K \le 288$. In Section 3.7, we present an alternative argument that yields the bound $K \le 1.783$ as stated in Theorem 3.5.1.

Before we pass to the argument, let us make one simple observation.

Exercise 3.5.2. ☕

(a) Check that the assumption of Grothendieck's inequality can be equivalently stated as follows:

$$\Big| \sum_{i,j} a_{ij} x_i y_j \Big| \le \max_i |x_i| \cdot \max_j |y_j|. \tag{3.12}$$

for any real numbers x_i and y_j.

(b) Show that the conclusion of Grothendieck's inequality can be equivalently stated as follows:

$$\Big| \sum_{i,j} a_{ij} \langle u_i, v_j \rangle \Big| \le K \max_i \|u_i\| \cdot \max_j \|v_j\| \tag{3.13}$$

for any Hilbert space H and any vectors $u_i, v_j \in H$.

Proof of Theorem 3.5.1 with $K \le 288$. **Step 1: Reductions.** Note that Grothendieck's inequality becomes trivial if we allow the value of K depend on the matrix $A = (a_{ij})$. (For example, $K = \sum_{ij} |a_{ij}|$ would work – check!) Let us choose $K = K(A)$ to be the *smallest* number that makes the conclusion (3.13) valid for a given matrix A and any Hilbert space H and any vectors $u_i, v_j \in H$. Our goal is to show that K does *not* depend on the matrix A or the dimensions m and n.

Without loss of generality,[4] we may do this for a specific Hilbert space H, namely for $\mathbb{R}^N$ equipped with the Euclidean norm $\|\cdot\|_2$. Let us fix vectors $u_i, v_j \in \mathbb{R}^N$ which realize the smallest K, that is

$$\sum_{i,j} a_{ij} \langle u_i, v_j \rangle = K, \quad \|u_i\|_2 = \|v_j\|_2 = 1.$$

Step 2: Introducing randomness. The main idea of the proof is to realize the vectors u_i, v_j via Gaussian random variables

$$U_i := \langle g, u_i \rangle \quad \text{and} \quad V_j := \langle g, v_j \rangle, \quad \text{where } g \sim N(0, I_N).$$

As we noted in Exercise 3.3.5, U_i and V_j are standard normal random variables whose correlations follow exactly the inner products of the vectors u_i and v_j:

$$\mathbb{E}\, U_i V_j = \langle u_i, v_j \rangle.$$

Thus

$$K = \sum_{i,j} a_{ij} \langle u_i, v_j \rangle = \mathbb{E} \sum_{i,j} a_{ij} U_i V_j. \tag{3.14}$$

Assume for a moment that the random variables U_i and V_j were bounded almost surely by some constant – say, by R. Then the assumption (3.12) of Grothendieck's inequality (after rescaling) would yield $\left| \sum_{i,j} a_{ij} U_i V_j \right| \le R^2$ almost surely, and (3.14) would then give $K \le R^2$.

Step 3: Truncation. Of course, this reasoning is flawed: the random variables $U_i, V_j \sim N(0,1)$ are not bounded almost surely. To fix this argument, we can utilize a useful *truncation* trick. Let us fix some level $R \ge 1$ and decompose the random variables as follows:

$$U_i = U_i^- + U_i^+ \quad \text{where} \quad U_i^- = U_i \, \mathbf{1}_{\{|U_i| \le R\}} \quad \text{and} \quad U_i^+ = U_i \, \mathbf{1}_{\{|U_i| > R\}}.$$

We similarly decompose $V_j = V_j^- + V_j^+$. Now U_i^- and V_j^- are bounded by R almost surely as we desired. The remainder terms U_i^+ and V_j^+ are small in the L^2 norm: indeed, the bound in Exercise 2.1.4 gives

$$\|U_i^+\|_{L^2}^2 \le 2\left(R + \frac{1}{R}\right) \frac{1}{\sqrt{2\pi}} e^{-R^2/2} < \frac{4}{R^2}, \tag{3.15}$$

[4] To see this, we can first trivially replace H with the subspace of H spanned by the vectors u_i and v_j (and with the norm inherited from H). This subspace has dimension at most $N := m + n$. Next, we recall the basic fact that all N-dimensional Hilbert spaces are isometric with each other, and in particular they are isometric to $\mathbb{R}^N$ with the norm $\|\cdot\|_2$. The isometry can be constructed by identifying orthogonal bases of those spaces.

and similarly for V_j^+.

Step 4: Breaking up the sum. The sum in (3.14) becomes

$$K = \mathbb{E} \sum_{i,j} a_{ij} (U_i^- + U_i^+)(V_j^- + V_j^+).$$

When we expand the product in each term we obtain four sums, which we proceed to bound individually. The first sum,

$$S_1 := \mathbb{E} \sum_{i,j} a_{ij} U_i^- V_j^-,$$

is the best of all. By construction, the random variables U_i^- and V_j^- are bounded almost surely by R. Thus, just like we explained above, we can use the assumption (3.12) of Grothendieck's inequality to get $S_1 \le R^2$.

We are not able to use the same reasoning for the second sum,

$$S_2 := \mathbb{E} \sum_{i,j} a_{ij} U_i^+ V_j^-,$$

since the random variable U_i^+ is unbounded. Instead, we will view the random variables U_i^+ and V_j^- as elements of the Hilbert space L^2 with the inner product $\langle X, Y \rangle_{L^2} = \mathbb{E}\, XY$. The second sum becomes

$$S_2 = \sum_{i,j} a_{ij} \left\langle U_i^+, V_j^- \right\rangle_{L^2}. \tag{3.16}$$

Recall from (3.15) that $\|U_i^+\|_{L^2} < 2/R$ and $\|V_j^-\|_{L^2} \le \|V_j\|_{L^2} = 1$ by construction. Then, applying the conclusion (3.13) of Grothendieck's inequality for the Hilbert space $H = L^2$, we find that[5]

$$S_2 \le K \cdot \frac{2}{R}.$$

The third and fourth sums, $S_3 := \mathbb{E} \sum_{i,j} a_{ij} U_i^- V_j^+$ and $S_4 := \mathbb{E} \sum_{i,j} a_{ij} U_i^+ V_j^+$, can be both bounded just like S_2. (Check!)

Step 5: Putting everything together. Putting the four sums together, we conclude from (3.14) that

$$K \le R^2 + \frac{6K}{R}.$$

Choosing $R = 12$ (for example) and solve the resulting inequality, we obtain $K \le 288$. The theorem is proved. □

Exercise 3.5.3 (Symmetric matrices, $x_i = y_i$)**.** ☕☕☕ Deduce the following version of Grothendieck's inequality for symmetric $n \times n$ matrices $A = (a_{ij})$ with

[5] It might seem weird that we are able to apply the inequality that we are trying to prove. Remember, however, that we chose K in the beginning of the proof as the best number that makes Grothendieck's inequality valid. This is the K we are using here.

real entries. Suppose that A is either positive semidefinite or has zero diagonal. Assume that, for any numbers $x_i \in \{-1, 1\}$, we have

$$\Big| \sum_{i,j} a_{ij} x_i x_j \Big| \le 1.$$

Then, for any Hilbert space H and any vectors $u_i, v_j \in H$ satisfying $\|u_i\| = \|v_j\| = 1$, we have

$$\Big| \sum_{i,j} a_{ij} \langle u_i, v_j \rangle \Big| \le 2K, \tag{3.17}$$

where K is the absolute constant from Grothendieck's inequality.

Hint: Check and use the polarization identity $\langle Ax, y \rangle = \langle Au, u \rangle - \langle Av, v \rangle$ where $u = (x + y)/2$ and $v = (x - y)/2$.

3.5.1 Semidefinite programming

One application area where Grothendieck's inequality can be particularly helpful is the analysis of certain computationally hard problems. A powerful approach to such problems is to try and *relax* them to computationally simpler and more tractable problems. This is often done using semidefinite programming, with Grothendieck's inequality guaranteeing the quality of such relaxations.

Definition 3.5.4. A *semidefinite program* is an optimization problem of the following type:

$$\text{maximize } \langle A, X \rangle : \quad X \succeq 0, \quad \langle B_i, X \rangle = b_i \text{ for } i = 1, \ldots, m. \tag{3.18}$$

Here A and B_i are given $n \times n$ matrices and b_i are given real numbers. The running "variable" X is an $n \times n$ symmetric positive semidefinite matrix, indicated by the notation $X \succeq 0$. The inner product

$$\langle A, X \rangle = \operatorname{tr}(A^{\mathsf{T}} X) = \sum_{i,j=1}^{n} A_{ij} X_{ij} \tag{3.19}$$

is the canonical inner product on the space of $n \times n$ matrices.

Note in passing that if we *minimize* instead of maximize in (3.18), we still get a semidefinite program. (To see this, replace A with $-A$.)

Every semidefinite program is a *convex program*, which maximizes a linear function $\langle A, X \rangle$ over a convex set of matrices. Indeed, the set of positive semidefinite matrices is convex (why?), and so is its intersection with the linear subspace defined by the constraints $\langle B_i, X \rangle = b_i$.

This is good news since convex programs are generally algorithmically tractable. There is a variety of computationally efficient solvers available for general convex programs, and for semidefinite programs (3.18) in particular, for example interior point methods.

Semidefinite relaxations

Semidefinite programs can be designed to provide computationally efficient relaxations of computationally hard problems, such as this one:

$$\text{maximize} \sum_{i,j=1}^{n} A_{ij} x_i x_j : \quad x_i = \pm 1 \text{ for } i = 1, \dots, n \tag{3.20}$$

where A is a given $n \times n$ symmetric matrix. This is an *integer optimization problem*. The feasible set consists of 2^n vectors $x = (x_i) \in \{-1, 1\}^n$, so finding the maximum by exhaustive search would take exponential time. Is there a smarter way to solve the problem? This is not likely: the problem (3.20) is known to be computationally hard in general (NP-hard).

Nonetheless, we can "relax" the problem (3.20) to a semidefinite program that can compute the maximum *approximately*, up to a constant factor. To formulate such a relaxation, let us replace in (3.20) the numbers $x_i = \pm 1$ by their higher-dimensional analogs – unit vectors X_i in $\mathbb{R}^n$. Thus we consider the following optimization problem:

$$\text{maximize} \sum_{i,j=1}^{n} A_{ij} \langle X_i, X_j \rangle : \quad \|X_i\|_2 = 1 \text{ for } i = 1, \dots, n. \tag{3.21}$$

Exercise 3.5.5. ☕☕ Show that the optimization (3.21) is equivalent to the following semidefinite program:

$$\text{maximize } \langle A, X \rangle : \quad X \succeq 0, \quad X_{ii} = 1 \text{ for } i = 1, \dots, n. \tag{3.22}$$

Hint: Consider the *Gram matrix* of the vectors X_i, which is the $n \times n$ matrix with entries $\langle X_i, X_j \rangle$. Do not forget to describe how to translate a solution of (3.22) into a solution of (3.21).

The guarantee of relaxation

We now see how Grothendieck's inequality guarantees the accuracy of semidefinite relaxations: the semidefinite program (3.21) approximates the maximum value in the integer optimization problem (3.20) up to an absolute constant factor.

Theorem 3.5.6. *Consider an $n \times n$ symmetric, positive semidefinite matrix A. Let $\mathrm{INT}(A)$ denote the maximum in the integer optimization problem (3.20) and $\mathrm{SDP}(A)$ denote the maximum in the semidefinite problem (3.21). Then*

$$\mathrm{INT}(A) \le \mathrm{SDP}(A) \le 2K \cdot \mathrm{INT}(A)$$

where $K \le 1.783$ is the constant in Grothendieck's inequality.

Proof The first bound follows with $X_i = (x_i, 0, 0, \dots, 0)^\mathsf{T}$. The second bound follows from Grothendieck's inequality for symmetric matrices in Exercise 3.5.3. (Argue that one can drop absolute values.) □

Although Theorem 3.5.6 allows us to approximate the maximum value in in (3.20), it is not obvious how to compute x_i's that attain this approximate value. Can we translate the vectors (X_i) that give a solution of the semidefinite program

(3.21) into labels $x_i = \pm 1$ that approximately solve (3.20)? In the next section, we illustrate this on the example of a remarkable NP-hard problem on graphs – the maximum cut problem.

Exercise 3.5.7. ☕☕☕ Let A be an $m \times n$ matrix. Consider the optimization problem

$$\text{maximize } \sum_{i,j} A_{ij} \left\langle X_i, Y_j \right\rangle : \quad \|X_i\|_2 = \|Y_j\|_2 = 1 \text{ for all } i, j$$

over $X_i, Y_j \in \mathbb{R}^k$ and $k \in \mathbb{N}$. Formulate this problem as a semidefinite program.
Hint: First, express the objective function as $\frac{1}{2}\operatorname{tr}(\tilde{A}ZZ^\mathsf{T})$, where $\tilde{A} = \left[\begin{smallmatrix} 0 & A \\ A^\mathsf{T} & 0 \end{smallmatrix}\right]$, $Z = \left[\begin{smallmatrix} X \\ Y \end{smallmatrix}\right]$ and X and Y are the matrices with rows X_i^T and Y_j^T, respectively. Then express the set of matrices of the type ZZ^T with unit rows as the set of symmetric positive semidefinite matrices whose diagonal entries equal 1.

3.6 Application: Maximum cut for graphs

We now illustrate the utility of semidefinite relaxations for the problem of finding the *maximum cut* of a graph, which is one of the well known NP-hard problems discussed in the computer science literature.

3.6.1 Graphs and cuts

An undirected *graph* $G = (V, E)$ is defined as a set V of vertices together with a set E of edges; each edge is an unordered pair of vertices. Here we consider finite, *simple* graphs – those with finitely many vertices and with no loops or multiple edges.

Definition 3.6.1 (Maximum cut)**.** Suppose we partition the set of vertices of a graph G into two disjoint sets. The *cut is the number of edges crossing between these two sets. The* maximum cut of G, denoted MAX-CUT(G), is obtained by maximizing the cut over all partitions of vertices; see Figure 3.10 for illustration.

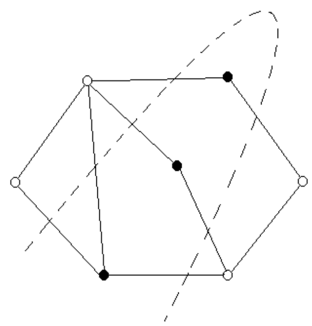

Figure 3.10 The dashed line illustrates the maximum cut of this graph, obtained by partitioning the vertices into the black and white ones. Here MAX-CUT$(G) = 7$.

Computing the maximum cut of a given graph is known to be a computationally hard problem (NP-hard).

3.6.2 A simple 0.5-approximation algorithm

We try to relax the maximum cut problem to a semidefinite program following the method we introduced in Section 3.5.1. To do this, we need to translate the problem into the language of linear algebra.

Definition 3.6.2 (Adjacency matrix)**.** The *adjacency matrix* A of a graph G on n vertices is a symmetric $n \times n$ matrix whose entries are defined as $A_{ij} = 1$ if the vertices i and j are connected by an edge and $A_{ij} = 0$ otherwise.

Let us label the vertices of G by the integers $1, \ldots, n$. A partition of the vertices into two sets can be described using a vector of labels

$$x = (x_i) \in \{-1, 1\}^n,$$

the sign of x_i indicating which subset the vertex i belongs to. For example, the three black vertices in Figure 3.10 may have labels $x_i = 1$, and the four white vertices labels $x_i = -1$. The cut of G corresponding to the partition given by x is simply the number of edges between the vertices with labels of opposite signs, i.e.

$$\mathrm{CUT}(G, x) = \frac{1}{2} \sum_{i,j:\, x_i x_j = -1} A_{ij} = \frac{1}{4} \sum_{i,j=1}^{n} A_{ij}(1 - x_i x_j). \tag{3.23}$$

(The factor $\frac{1}{2}$ prevents double counting of edges (i, j) and (j, i).) The maximum cut is then obtained by maximizing $\mathrm{CUT}(G, x)$ over all x, that is

$$\text{MAX-CUT}(G) = \frac{1}{4} \max \left\{ \sum_{i,j=1}^{n} A_{ij}(1 - x_i x_j) :\ x_i = \pm 1 \text{ for all } i \right\}. \tag{3.24}$$

Let us start with a simple 0.5-approximation algorithm for maximum cut – one which finds a cut with at least *half* of the edges of G.

Proposition 3.6.3 (0.5-approximation algorithm for maximum cut)**.** *Partition the vertices of G into two sets at random, uniformly over all 2^n partitions. Then the expectation of the resulting cut equals*

$$0.5|E| \geq 0.5\, \text{MAX-CUT}(G),$$

where $|E|$ denotes the total number of edges of G.

Proof The random cut is generated by a symmetric Bernoulli random vector $x \sim \mathrm{Unif}\left(\{-1, 1\}^n\right)$, which has independent symmetric Bernoulli coordinates. Then, in (3.23) we have $\mathbb{E}\, x_i x_j = 0$ for $i \neq j$ and $A_{ij} = 0$ for $i = j$ (since the graph has no loops). Thus, using linearity of expectation, we get

$$\mathbb{E}\, \mathrm{CUT}(G, x) = \frac{1}{4} \sum_{i,j=1}^{n} A_{ij} = \frac{1}{2}|E|.$$

This completes the proof. □

Exercise 3.6.4. ☕☕ For any $\varepsilon > 0$, give an $(0.5 - \varepsilon)$-approximation algorithm for maximum cut, which is always *guaranteed* to give a suitable cut, but may have a random running time. Give a bound on the expected running time.

Hint: Consider cutting G repeatedly. Give a bound on the expected number of experiments.

3.6.3 Semidefinite relaxation

Now we will do much better and give a 0.878-approximation algorithm, which is due to Goemans and Williamson. It is based on a semidefinite relaxation of the NP-hard problem (3.24). It should be easy to guess what such relaxation could be: recalling (3.21), it is natural to consider the semidefinite problem

$$\mathrm{SDP}(G) := \frac{1}{4} \max \left\{ \sum_{i,j=1}^{n} A_{ij}(1 - \langle X_i, X_j \rangle) : \ X_i \in \mathbb{R}^n, \ \|X_i\|_2 = 1 \text{ for all } i \right\}. \tag{3.25}$$

(Again – why is this a semidefinite program?)

As we will see, not only the value $\mathrm{SDP}(G)$ approximates $\text{MAX-CUT}(G)$ to within the 0.878 factor, but we can obtain an actual partition of G (i.e., the labels x_i) which attains this value. To do this, we describe how to translate a solution (X_i) of (3.25) into labels $x_i = \pm 1$.

This can be done by the following *randomized rounding* step. Choose a random hyperplane in $\mathbb{R}^n$ passing through the origin. It cuts the set of vectors X_i into two parts; let us assign labels $x_i = 1$ to one part and $x_i = -1$ to the other part. Equivalently, we may choose a standard normal random vector

$$g \sim N(0, I_n)$$

and define

$$x_i := \operatorname{sign} \langle X_i, g \rangle, \quad i = 1, \ldots, n. \tag{3.26}$$

See Figure 3.11 for an illustration.[6]

Theorem 3.6.5 (0.878-approximation algorithm for maximum cut)**.** *Let G be a graph with adjacency matrix A. Let $x = (x_i)$ be the result of a randomized rounding of the solution (X_i) of the semidefinite program (3.25). Then*

$$\mathbb{E}\,\mathrm{CUT}(G, x) \geq 0.878\,\mathrm{SDP}(G) \geq 0.878\,\text{MAX-CUT}(G).$$

The proof of this theorem will be based on the following elementary identity. We can think of it as a more advanced version of the identity (3.6), which we used in the proof of Grothendieck's inequality, Theorem 3.5.1.

[6] In the rounding step, instead of the normal distribution we could use any other rotation invariant distribution in $\mathbb{R}^n$, for example the uniform distribution on the sphere S^{n-1}.

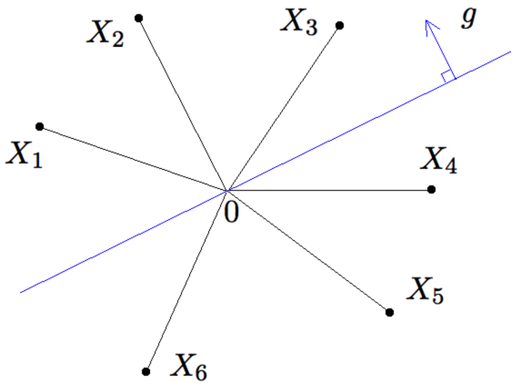

Figure 3.11 Randomized rounding of vectors $X_i \in \mathbb{R}^n$ into labels $x_i = \pm 1$. For this configuration of points X_i and a random hyperplane with normal vector g, we assign $x_1 = x_2 = x_3 = 1$ and $x_4 = x_5 = x_6 = -1$.

Lemma 3.6.6 (Grothendieck's identity)**.** *Consider a random vector $g \sim N(0, I_n)$. Then, for any fixed vectors $u, v \in S^{n-1}$, we have*

$$\mathbb{E}\,\mathrm{sign}\,\langle g, u\rangle\,\mathrm{sign}\,\langle g, v\rangle = \frac{2}{\pi}\arcsin\langle u, v\rangle\,.$$

Exercise 3.6.7. ☕☕ Prove Grothendieck's identity.

Hint: It will quickly follow once you show that the probability that $\langle g, u\rangle$ and $\langle g, v\rangle$ have opposite signs equals α/π, where $\alpha \in [0, \pi]$ is the angle between the vectors u and v. To check this, use rotation invariance to reduce the problem to $\mathbb{R}^2$. Once on the plane, rotation invariance will give the result.

A weak point of Grothendieck's identity is the non-linear function arcsin, which would be hard to work with. Let us replace it with a linear function using the numeric inequality

$$1 - \frac{2}{\pi}\arcsin t = \frac{2}{\pi}\arccos t \ge 0.878(1-t), \quad t \in [-1, 1], \tag{3.27}$$

which can be easily verified using software; see Figure 3.12.

Proof of Theorem 3.6.5 By (3.23) and linearity of expectation, we have

$$\mathbb{E}\,\mathrm{CUT}(G, x) = \frac{1}{4}\sum_{i,j=1}^{n} A_{ij}(1 - \mathbb{E}\,x_i x_j).$$

The definition of labels x_i in the rounding step (3.26) gives

$$\begin{aligned} 1 - \mathbb{E}\,x_i x_j &= 1 - \mathbb{E}\,\mathrm{sign}\,\langle X_i, g\rangle\,\mathrm{sign}\,\langle X_j, g\rangle \\ &= 1 - \frac{2}{\pi}\arcsin\langle X_i, X_j\rangle \quad \text{(by Grothendieck's identity, Lemma 3.6.6)} \\ &\ge 0.878(1 - \langle X_i, X_j\rangle) \quad \text{(by (3.27))}. \end{aligned}$$

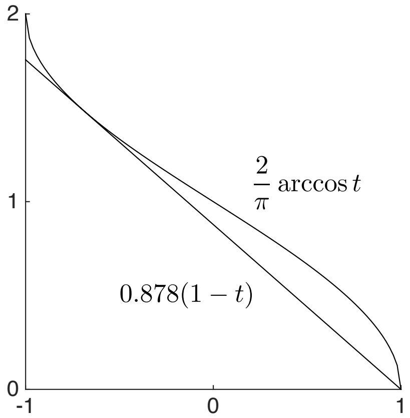

Figure 3.12 The inequality $\frac{2}{\pi}\arccos t \geq 0.878(1-t)$ holds for all $t \in [-1,1]$.

Therefore

$$\mathbb{E}\,\mathrm{CUT}(G,x) \geq 0.878 \cdot \frac{1}{4}\sum_{i,j=1}^{n} A_{ij}(1-\langle X_i, X_j\rangle) = 0.878\,\mathrm{SDP}(G).$$

This proves the first inequality in the theorem. The second inequality is trivial since $\mathrm{SDP}(G) \geq \text{MAX-CUT}(G)$. (Why?) □

3.7 Kernel trick, and tightening of Grothendieck's inequality

Our proof of Grothendieck's inequality given in Section 3.5 yields a very loose bound on the absolute constant K. We now give an alternative proof that gives (almost) the best known constant $K \leq 1.783$.

Our new argument will be based on Grothendieck's identity (Lemma (3.6.6)). The main challenge in using this identity arises from the non-linearity of the function $\arcsin(x)$. Indeed, suppose there were no such nonlinearity, and we hypothetically had $\mathbb{E}\,\mathrm{sign}\,\langle g,u\rangle\,\mathrm{sign}\,\langle g,v\rangle = \frac{2}{\pi}\langle u,v\rangle$. Then Grothendieck's inequality would easily follow:

$$\frac{2}{\pi}\sum_{i,j} a_{ij}\,\langle u_i, v_j\rangle = \sum_{i,j} a_{ij}\,\mathbb{E}\,\mathrm{sign}\,\langle g,u_i\rangle\,\mathrm{sign}\,\langle g,v_j\rangle \leq 1,$$

where in the last step we swapped the sum and expectation and used the assumption of Grothendieck's inequality with $x_i = \mathrm{sign}\,\langle g,u_i\rangle$ and $y_j = \mathrm{sign}\,\langle g,y_j\rangle$. This would give Grothendieck's inequality with $K \leq \pi/2 \approx 1.57$.

This argument is of course wrong. To address the non-linear form $\frac{2}{\pi}\arcsin\langle u,v\rangle$ that appears in Grothendieck's identity, we use the following remarkably powerful trick: represent $\frac{2}{\pi}\arcsin\langle u,v\rangle$ as the (linear) inner product $\langle u',v'\rangle$ of some other vectors u', v' in some Hilbert space H. In the literature on machine learning, this method is called the *kernel trick*.

We will explicitly construct the non-linear transformations $u' = \Phi(u)$, $v' = \Psi(v)$ that will do the job. Our construction is convenient to describe in the language of *tensors*, which are a higher dimensional generalization of the notion of matrices.

Definition 3.7.1 (Tensors)**.** A tensor can be described as a multidimensional array. Thus, a k-th order tensor $(a_{i_1 \ldots i_k})$ is a k-dimensional array of real numbers $a_{i_1 \ldots i_k}$. The canonical inner product on $\mathbb{R}^{n_1 \times \cdots \times n_k}$ defines the inner product of tensors $A = (a_{i_1 \ldots i_k})$ and $B = (b_{i_1 \ldots i_k})$:

$$\langle A, B \rangle := \sum_{i_1, \ldots, i_k} a_{i_1 \ldots i_k} b_{i_1 \ldots i_k}. \tag{3.28}$$

Example 3.7.2. Scalars, vectors and matrices are examples of tensors. As we noted in (3.19), for $m \times n$ matrices the inner product of tensors (3.28) specializes to

$$\langle A, B \rangle = \mathrm{tr}(A^\mathsf{T} B) = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} B_{ij}.$$

Example 3.7.3 (Rank-one tensors)**.** Every vector $u \in \mathbb{R}^n$ defines the k-th order *tensor product* $u \otimes \cdots \otimes u$, which is the tensor whose entries are the products of all k-tuples of the entries of u. In other words,

$$u \otimes \cdots \otimes u = u^{\otimes k} := (u_{i_1} \cdots u_{i_k}) \in \mathbb{R}^{n \times \cdots \times n}.$$

In particular, for $k = 2$, the tensor product $u \otimes u$ is just the $n \times n$ matrix which is the outer product of u with itself:

$$u \otimes u = (u_i u_j)_{i,j=1}^{n} = u u^\mathsf{T}.$$

One can similarly define the tensor products $u \otimes v \otimes \cdots \otimes z$ for different vectors $u, v, \ldots, z$.

Exercise 3.7.4. ☕ Show that for any vectors $u, v \in \mathbb{R}^n$ and $k \in \mathbb{N}$, we have

$$\left\langle u^{\otimes k}, v^{\otimes k} \right\rangle = \langle u, v \rangle^k.$$

This exercise shows a remarkable fact: we can represent non-linear forms like $\langle u, v \rangle^k$ as the usual, *linear* inner product in some other space. Formally, there exist a Hilbert space H and a transformation $\Phi : \mathbb{R}^n \to H$ such that

$$\big\langle \Phi(u), \Phi(v) \big\rangle = \langle u, v \rangle^k.$$

In this case, H is the space of k-th order tensors, and $\Phi(u) = u^{\otimes k}$.

In the next two exercises, we extend this observation to more general non-linearities.

Exercise 3.7.5. ☕☕

(a) Show that there exist a Hilbert space H and a transformation $\Phi : \mathbb{R}^n \to H$ such that

$$\big\langle \Phi(u), \Phi(v) \big\rangle = 2 \langle u, v \rangle^2 + 5 \langle u, v \rangle^3 \quad \text{for all } u, v \in \mathbb{R}^n.$$

Hint: Consider the cartesian product $H = \mathbb{R}^{n \times n} \oplus \mathbb{R}^{n \times n \times n}$.

(b) More generally, consider a polynomial $f : \mathbb{R} \to \mathbb{R}$ with non-negative coefficients, and construct H and Φ such that

$$\big\langle \Phi(u), \Phi(v) \big\rangle = f(\langle u, v \rangle) \quad \text{for all } u, v \in \mathbb{R}^n.$$

(c) Show the same for any *real analytic function* $f : \mathbb{R} \to \mathbb{R}$ with non-negative coefficients, i.e. for any function that can be represented as a convergent series

$$f(x) = \sum_{k=0}^{\infty} a_k x^k, \quad x \in \mathbb{R}, \tag{3.29}$$

and such that $a_k \geq 0$ for all k.

Exercise 3.7.6. ☕ Let $f : \mathbb{R} \to \mathbb{R}$ be any real analytic function (with possibly negative coefficients in (3.29)). Show that there exist a Hilbert space H and transformations $\Phi, \Psi : \mathbb{R}^n \to H$ such that

$$\big\langle \Phi(u), \Psi(v) \big\rangle = f(\langle u, v \rangle) \quad \text{for all } u, v \in \mathbb{R}^n.$$

Moreover, check that

$$\|\Phi(u)\|^2 = \|\Psi(u)\|^2 = \sum_{k=0}^{\infty} |a_k| \|u\|_2^{2k}.$$

Hint: Construct Φ as in Exercise 3.7.5 with Φ, but include the signs of a_k in the definition of Ψ.

Let us specialize the kernel trick to the non-linearity $\frac{2}{\pi} \arcsin \langle u, v \rangle$ that appears in Grothendieck's identity.

Lemma 3.7.7. *There exists a Hilbert space H and transformations*[7] $\Phi, \Psi : S^{n-1} \to S(H)$ *such that*

$$\frac{2}{\pi} \arcsin \big\langle \Phi(u), \Psi(v) \big\rangle = \beta \langle u, v \rangle \quad \textit{for all } u, v \in S^{n-1}, \tag{3.30}$$

where $\beta = \frac{2}{\pi} \ln(1 + \sqrt{2})$.

Proof Rewrite the desired identity (3.30) as

$$\big\langle \Phi(u), \Psi(v) \big\rangle = \sin\left(\frac{\beta\pi}{2} \langle u, v \rangle \right). \tag{3.31}$$

The result of Exercise 3.7.6 gives us the Hilbert space H and the maps Φ, Ψ :

[7] Here $S(H)$ denotes the unit sphere of the Hilbert space H.

$\mathbb{R}^n \to H$ that satisfy (3.31). It only remains to determine the value of β for which Φ and Ψ map unit vectors to unit vectors. To do this, we recall the Taylor series

$$\sin t = t - \frac{t^3}{3!} + \frac{t^5}{5!} - \cdots \quad \text{and} \quad \sinh t = t + \frac{t^3}{3!} + \frac{t^5}{5!} + \cdots$$

Exercise 3.7.6 then guarantees that for every $u \in S^{n-1}$, we have

$$\|\Phi(u)\|^2 = \|\Psi(u)\|^2 = \sinh\left(\frac{\beta\pi}{2}\right).$$

This quantity equals 1 if we set

$$\beta := \frac{2}{\pi} \operatorname{arcsinh}(1) = \frac{2}{\pi} \ln(1+\sqrt{2}).$$

The lemma is proved. □

Now we are ready to prove Grothendieck's inequality (Theorem 3.5.1) with constant

$$K \le \frac{1}{\beta} = \frac{\pi}{2\ln(1+\sqrt{2})} \approx 1.783.$$

Proof of Theorem 3.5.1 We can assume without loss of generality that $u_i, v_j \in S^{N-1}$ (this is the same reduction as we did in the proof in Section 3.5). Lemma 3.7.7 gives us unit vectors $u_i' = \Phi(u_i)$ and $v_j' = \Psi(v_j)$ in some Hilbert space H, which satisfy

$$\frac{2}{\pi} \arcsin \left\langle u_i', v_j' \right\rangle = \beta \left\langle u_i, v_j \right\rangle \quad \text{for all } i, j.$$

We can again assume without loss of generality that $H = \mathbb{R}^M$ for some M. (Why?) Then

$$\begin{aligned}
\beta \sum_{i,j} a_{ij} \left\langle u_i, v_j \right\rangle &= \sum_{i,j} a_{ij} \cdot \frac{2}{\pi} \arcsin \left\langle u_i', v_j' \right\rangle \\
&= \sum_{i,j} a_{ij}\, \mathbb{E} \operatorname{sign} \langle g, u_i' \rangle \operatorname{sign} \left\langle g, v_j' \right\rangle \quad \text{(by Lemma 3.6.6)}, \\
&\le 1,
\end{aligned}$$

where in the last step we swapped the sum and expectation and used the assumption of Grothendieck's inequality with $x_i = \operatorname{sign} \langle g, u_i' \rangle$ and $y_j = \operatorname{sign} \left\langle g, y_j' \right\rangle$. This yields the conclusion of Grothendieck's inequality for $K \le 1/\beta$. □

3.7.1 Kernels and feature maps

Since the kernel trick was so successful in the proof of Grothendieck's inequality, we may ask – what other non-linearities can be handled with the kernel trick? Let

$$K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$$

be a function of two variables on a set $\mathcal{X}$. Under what conditions on K can we find a Hilbert space H and a transformation

$$\Phi : \mathcal{X} \to H$$

so that

$$\big\langle \Phi(u), \Phi(v) \big\rangle = K(u,v) \quad \text{for all } u, v \in \mathcal{X}? \tag{3.32}$$

The answer to this question is provided by Mercer's and, more precisely, Moore-Aronszajn's theorems. The necessary and sufficient condition is that K be a *positive semidefinite kernel*, which means that for any finite collection of points $u_1, \ldots, u_N \in \mathcal{X}$, the matrix

$$\Big(K(u_i, u_j) \Big)_{i,j=1}^{N}$$

is symmetric and positive semidefinite. The map Φ is called a *feature map*, and the Hilbert space H can be constructed from the kernel K as a (unique) *reproducing kernel Hilbert space.*

Examples of positive semidefinite kernels on $\mathbb{R}^n$ that are common in machine learning include the *Gaussian kernel* (also called the radial basis function kernel)

$$K(u,v) = \exp\Big(-\frac{\|u-v\|_2^2}{2\sigma^2} \Big), \quad u, v \in \mathbb{R}^n,\ \sigma > 0$$

and the *polynomial kernel*

$$K(u,v) = \Big(\langle u, v \rangle + r \Big)^k, \quad u, v \in \mathbb{R}^n,\ r > 0,\ k \in \mathbb{N}.$$

The kernel trick (3.32), which represents a general kernel $K(u,v)$ as an inner product, is very popular in *machine learning.* It allows one to handle non-linear models (determined by kernels K) by using methods developed for linear models. In contrast to what we did in this section, in machine learning applications the explicit description of the Hilbert space H and the feature map $\Phi : \mathcal{X} \to H$ is typically not needed. Indeed, to compute the inner product $\big\langle \Phi(u), \Phi(v) \big\rangle$ in H, one does not need to know Φ: the identity (3.32) allows one to compute $K(u,v)$ instead.

3.8 Notes

Theorem 3.1.1 about the concentration of the norm of random vectors is known but difficult to locate in the existing literature. We will later prove a more general result, Theorem 6.3.2, which is valid for anisotropic random vectors. It is unknown if the quadratic dependence on K in Theorem 3.1.1 is optimal. One may also wonder about concentration of the norm $\|X\|_2$ of random vectors X whose coordinates are not necessarily independent. In particular, for a random vector X that is uniformly distributed in a convex set K, concentration of the norm is one of the central problems in geometric functional analysis; see [93, Section 2] and [36, Chapter 12].

Exercise 3.3.4 mentions Cramér-Wold's theorem. It a straightforward consequence of the uniqueness theorem for characteristic functions, see [23, Section 29].

The concept of frames introduced in Section 3.3.4 is an important extension of the notion of orthogonal bases. One can read more about frames and their applications in signal processing and data compression e.g. in [51, 121].

Sections 3.3.5 and 3.4.4 discuss random vectors uniformly distributed in convex sets. The books [11, 36] study this topic in detail, and the surveys [185, 218] discuss algorithmic aspects of computing the volume of convex sets in high dimensions.

Our discussion of sub-gaussian random vectors in Section 3.4. mostly follows [222]. An alternative geometric proof of Theorem 3.4.6 can be found in [13, Lemma 2.2].

Grothendieck's inequality (Theorem 3.5.1) was originally proved by A. Grothendieck in 1953 [90] with bound on the constant $K \le \sinh(\pi/2) \approx 2.30$; a version of this original argument is presented [133, Section 2]. There is a number of alternative proofs of Grothendieck's inequality with better and worse bounds on K; see [35] for the history. The surveys [115, 168] discuss ramifications and applications of Grothendieck's inequality in various areas of mathematics and computer science. Our first proof of Grothendieck's inequality, the one given in Section 3.5, is similar to the one in [5, Section 8.1]; it was kindly brought to author's attention by Mark Rudelson. Our second proof, the one from Section 3.7, is due to J.-L. Krivine [122]; versions of this argument can be found e.g. in [7] and [126]. The bound on the constant $K \le \frac{\pi}{2\ln(1+\sqrt{2})} \approx 1.783$ that follows from Krivine's argument is currently the best known *explicit* bound on K. It has been proved, however, that the best possible bound must be strictly smaller than Krivine's bound, but no explicit number is known [35].

A part of this chapter is about semidefinite relaxations of hard optimization problems. For an introduction to the area of convex optimization, including semidefinite programming, we refer to the books [34, 39, 126, 29]. For the use of Grothendieck's inequality in analyzing semidefinite relaxations, see [115, 7]. Our presentation of the maximum cut problem in Section 3.6 follows [39, Section 6.6] and [126, Chapter 7]. The semidefinite approach to maximum cut, which we discussed in Section 3.6.3, was pioneered in 1995 by M. Goemans and D. Williamson [83]. The approximation ratio $\frac{2}{\pi} \min_{0 \le \theta \le \pi} \frac{\theta}{1-\cos(\theta)} \approx 0.878$ guaranteed by the Goemans-Williamson algorithm remains the best known constant for the max-cut problem. If the Unique Games Conjecture is true, this ratio can not be improved, i.e. any better approximation would be NP-hard to compute [114].

In Section 3.7 we give Krivine's proof of Grothendieck's inequality [122]. We also briefly discuss kernel methods there. To learn more about kernels, reproducing kernel Hilbert spaces and their applications in machine learning, see e.g. the survey [102].